Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We consider the problem of covering/connecting a multi-interface network with a minimum cost assignment of interfaces to vertices. In the covering problem we demand that every edge is covered by at least one common interface assigned to its endpoints. In the connection problem we are given a set of terminals $U \subseteq V$ and we demand that the subgraph of activated edges spans all of them. We give an $O(\log m)$ approximation algorithm for both min-sum and min-max versions of the covering problem. The algorithm is based on rounding the optimal solution of a natural, set-cover-style linear programming relaxation of the problem. We also show how to use this LP to obtain a k-approximation algorithm for the min-sum and min-max covering problems, where k is the number of interfaces. Importantly, all of the aforementioned procedures work even when the cost of activating an interface depends on the interface as well as the vertex to which it is assigned. For min-max covering this improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation, where b is a certain parameter of the input instance which can be as large as $\Omega(n)$ and $c_{\max}$ is the maximum cost of activating an interface and $O(k)$-approximation for the uniform cost case [Algorithmica 2012]. For min-sum covering this improves the previously best known $O\left(\sqrt{n} \cdot \log n\right)$-approximation and generalizes the $O(k)$-approximation [Wireless Networks 2009]. For the connection problem we devise an $O\left(\log^2 m\right)$-approximation algorithm for both min-sum and min-max versions, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity of the terminals. We also give combinatorial algorithms with $O(\Delta)$ guarantees for both min-sum and min-max covering as well as min-sum connection thus improving on the previously known $O(\Delta \cdot c_{\max})$ for the min-max covering.

We complement these results by showing that both the min-sum covering and connecting cannot be approximated within a factor of $(1 - o(1)) \cdot \ln n$ even on stars, unless $P = NP$. We also show, by an approximation-preserving reduction to the min-sum covering assignment problem on bipartite graphs with unit costs, that min-sum covering cannot be approximated within a factor of $o(\log \Delta)$ unless $P = NP$.

2012 ACM Subject Classification Replace ccsdesc macro with valid one

Keywords and phrases Multi-Interface Networks, Approximation Algorithms, Covering, Connecting

Digital Object Identifier 10.4230/LIPIcs.CVIT.2016.23

1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by choosing between large variety of interfaces available at each device. Moreover, the devices present in the network may vary according to the interfaces they provide for the usage. For example some device may be able to connect by Bluetooth and Wi-Fi, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as one of two graph problems, depending on the type of the connection one wishes to establish. Each devices in the netowrk is modeled as vertex of a graph. Additionally, with each such vertex we associate a set of labels representing the available interfaces. In the covering problem each edge in this graph represents a connection that we wish to establish by assigning a common interface to its endpoints. To see an

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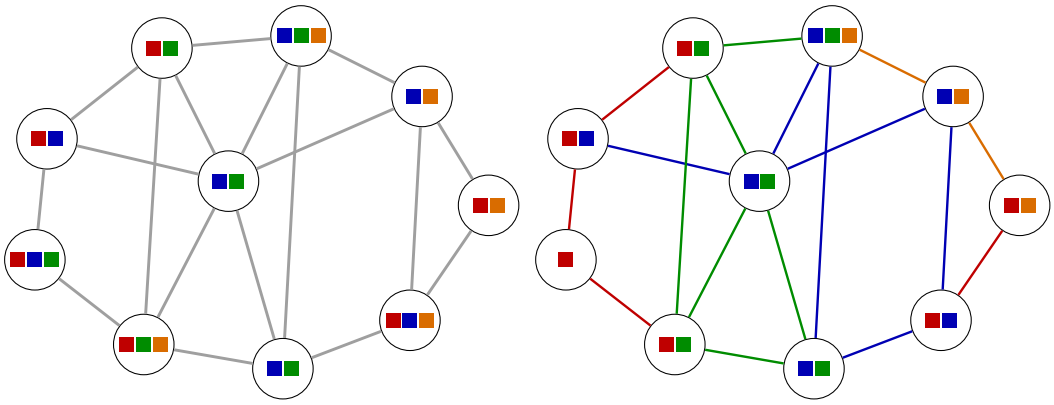
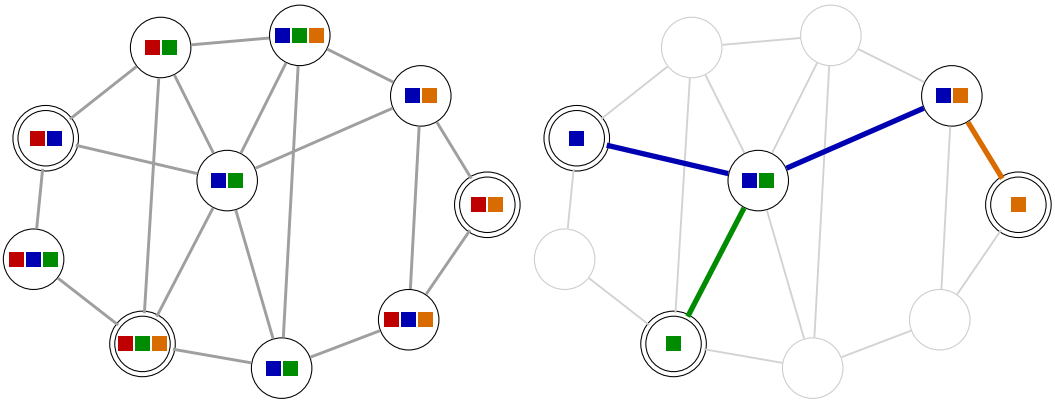
42nd Conference on Very Important Topics (CVIT 2016).
Editors: John Q. Open and Joan R. Access; Article No. 23; pp. 23:1–23:13
Leibniz International Proceedings in Informatics
LIPICS Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany



Available labels

Example covering assignment

Figure 1 An example instance of a multi-interface network with available interfaces at each vertex and a feasible covering assignment.



Available labels (double contour = terminal)

Example connecting assignment

Figure 2 An example instance of a multi-interface network with available interfaces at each vertex and a feasible connecting assignment.

example, see Figure 1. In the connection problem we are additionally given a set of terminals $U \subseteq V$; each edge can be activated by assigning a common interface to its endpoints, and we require that the subgraph of activated edges connects all terminals (vertices outside U may be used as Steiner vertices). To see an example, see Figure 2 In both problems, we assume that activating an interface at a given vertex incurs some arbitrary cost. The goal is to minimize either the min-max cost (the maximum cost per vertex) or min-sum cost (sum of costs over all vertices) of the interface assignment. Both problems are NP-hard and we wish to find good approximation algorithms.

1.1 Related Work

The algorithmic study of multi-interface networks goes back at least to the work of Klasing, Kosowski, and Navarra [14]. When k is bounded they show APX-hardness and provide an $\min\left\{\left\lceil \frac{k+1}{2}, \frac{2m}{n} \right\rceil\right\}$-approximation algorithm for the min-sum version of the covering problem with uniform costs. For unbounded k they give an $\min\{k-1, \sqrt{n}(1+\ln n)\}$-approximation for interface dependent costs and prove that the problem is NP-hard to approximate within

$O(\log k)$-factor. Assuming $U = V$, the min-sum connection problem for $k \geq 2$ is known to be APX-hard even for uniform costs and a $(2 - 1/k)$-approximation algorithm is known as well as 4/3-approximation for $k = 2$. Non uniform, interface dependent costs version admits 2-approximation [16]. The same research direction also led to results for other classical optimization problems in the multi-interface setting, including maximum matching [15] and flow optimization with bandwidth constraints [9].

The min-max coverage problem is known to be NP-hard for any fixed $\Delta \geq 5$ and $k \geq 16$ and cannot be approximatet within a factor of $\eta \ln \Delta$ unless P=NP even for uniform costs on trees (for a certain constant η). On the algorithmic side the problem admits an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation algorithm, where b denotes the existence of a b-bounded ownership function as well as $O\{k \cdot c_{max}\}$ and $O\{\Delta \cdot c_{max}\}$-approximation algorithms. Assuming $U = V$, the min-max connection is known to be NP-hard for any fixed $\Delta \geq 3$ and $k \geq 10$, but is solvable for $k \leq 2$. By equivalence, for the case of trees the problem admits $O(\log n)$-approximation (since for trees $b = 1$) and $O\{k \cdot c_{max}\}$ and $O\{\Delta \cdot c_{max}\}$-approximation. No such results are known for the general case [8]

Another active line of research concerns constrained and parameterized versions of coverage. In the *p-coverage* setting, where each node may activate at most p interfaces, the problem is NP-complete already for $p = 2$ on graphs with maximum degree $\Delta \geq 4$, even with uniform costs [1]. Aloisio and Navarra showed that for $p = 2$ the problem is FPT with respect to treewidth tw via Courcelle's theorem. They also gave an explicit dynamic programming algorithms, for carvingwidth and pathwidth [1]. The pathwidth case was subsequently revisited and extended in [2]. Most recently the covering problem was studied in [4], establishing FPT results including an algorithm parametrized by $k + \mathrm{tw}$ and hardness lower bounds with respect to several structural parameters, including vertex cover number, feedback edge number and distance to clique.

1.2 Our contribution and techniques

In this paper we give a first $O(\log m)$-approximation algorithm for all four variants (covering/connection and min-sum/min-max). This improves the previously known result of [8], which gave an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation for the min-max covering version, where b is the parameter related to the existence of a b-owner function [11, 7]. This als improves the $O(\sqrt{n} \cdot \log n)$-approximation for the min-sum covering problem by [14]. Our approach is based on a set-cover-style LP relaxation together with a randomized rounding. We also show how to use the same LP relaxation to obtain a k-approximation algorithm for the min-sum and min-max covering problems, where k is the number of interfaces. All these guarantees hold in the non-uniform setting where activation costs depend on both the interface and the vertex it is assigned to.

Then, we shift to combinatorial algorithms. We give a Δ-approximation algorithm for min-sum covering, a 1.39Δ-approximation algorithm for min-sum connection, and a 2Δ-approximation algorithm for min-max covering, where Δ is the maximum degree of the input graph. In particular, for min-max covering this improves on the previously known $O(\Delta \cdot c_{\max})$ approximation, where $c_{\max}$ is the maximum interface-activation cost.

The advantage of all of our algorithms is that they are simple and follow some kind of a greedy choice, such as randomized rounding or selecting the cheapest interface for a given edge. We complement these positive results with hardness: we show that both the min-sum covering and connecting problems cannot be approximated within a factor of $(1 - o(1)) \cdot \ln n$ even on stars, unless $P = NP$. We also prove that even the unit-cost min-sum covering assignment problem on bipartite graphs cannot be approximated within a factor of $o(\log \Delta)$

unless $P = NP$. Since analogous logarithmic hardness is known for min-max variants of the problems [8], our $O(\log m)$-approximation guarantees are tight up to constant factors for the case when Δ is unbounded.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $\lambda\colon V \to [k]$ be a function that assigns to each vertex $v \in V$ a set of available interfaces. In the *covering problem* we wish to find an assignment $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$ and for every edge $uv \in E$ we have $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$. We call such an assignment a *covering assignment*. In the *connection problem* we are additionally given a set of *terminals* $U \subseteq V$. We wish to find an assignment $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$ and the subgraph G_A of G induced by the edges $uv \in E$ with $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$ connects all terminals, i.e., every two vertices of U lie in the same connected component of G_A. Vertices in $V \setminus U$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $U = V$ corresponds to requiring G_A to be fully connected.

We additionally assume, that a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$ is given, which denotes the cost of assingining the interface $i \in [k]$ to vertex $v \in V$. The sum cost of the assignment is defined as:

$$\texttt{COST}_A = \sum_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-sum assignment problem is to find an assignment λ_A that minimizes $\texttt{COST}_A$. The max cost of the assignment is defined as:

$$\texttt{COST}_W = \max_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-max assignment problem is to find an assignment λ_A that minimizes $\texttt{COST}_W$.

We will make use of the following probability fact known as the Chernoff bound [6]:

▶ **Theorem 1** (Chernoff Bound). *Let $X_1, \ldots, X_n$ be independent random variables taking values in $\{0, 1\}$ and let $X = \sum_{i=1}^{n} X_i$. Then for every $\delta > 0$:*

$$\Pr[X \geq (1+\delta) \cdot \mathbb{E}[X]] \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}} \right)^{\mathbb{E}[X]}$$

and for every $0 < \delta < 1$:

$$\Pr[X \leq (1-\delta) \cdot \mathbb{E}[X]] \leq \left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}} \right)^{\mathbb{E}[X]}.$$

3 LP based approximation algorithms

Below we show an $O(\log m)$ approximation algorithms for all four variants and $O(k)$-approximation for the covering problems. The idea of our ILP formulation and the algorithm is inspired by the rounding procedure known for set cover [17] and related problems [3]

3.1 ILP formulation of covering multi-interface networks

We start with the following ILP formulation of the min-sum covering assignment problem. Let $x_{i,v}$ be a binary variable that is 1 if and only if the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. Then the min-sum covering assignment problem can be formulated as follows:

$$\begin{aligned} \min \quad & \sum_{v \in V} \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \\ \text{s.t.} \quad & \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E \\ & x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \end{aligned}$$

Similarly, the min-max covering assignment problem can be formulated as follows:

$$\begin{aligned} \min \quad & M \\ \text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\ & \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E \\ & x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \end{aligned}$$

Note that the above formulations are not linear due to the presence of the min function. However, by adding an additional variable $y_{i,uv}$ we can linearize the constraints of type $\sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1$ by replacing them with $\sum_{i \in \lambda(u) \cap \lambda(v)} y_{i,uv} \geq 1$ together with the constraints $y_{i,uv} \leq x_{i,u}$ and $y_{i,uv} \leq x_{i,v}$ for every $i \in \lambda(u) \cap \lambda(v)$ and $uv \in E$. Thus, we can solve the LP relaxation of each of the above formulations in polynomial time.

3.2 Randomized $O(\log m)$-approximation algorithm for the covering problem

Algorithm 1 A relaxation-based $O(\log m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}$.
2. Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
3. Set $\lambda_A(v) = \{i \in \lambda(v) \colon 3 \ln m \cdot x_{i,v} \geq t_i\}$, for every $v \in V$.
4. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 2.** *The above algorithm is an $O(\log m)$-approximation algorithm for both the min-sum and the min-max covering assignment problems which succeeds with high probability.*

Proof. We have the following lemmas:

▶ **Lemma 3.** *The algorithm returns a solution of cost at most $9 \ln m \cdot \mathtt{OPT}$ with high probability.*

Proof. By $X_{v,i}$ denote the random variable that is 1 if $3 \ln m \cdot x_{i,v} \geq t_i$ and 0 otherwise. Then we can write $c(\lambda_A(v)) = \sum_{i \in \lambda(v)} X_{v,i} \cdot c(i,v)$. We have: $\mathbb{E}[X_{v,i}] = \Pr[i \in \lambda_A(v)] = 3 \ln m \cdot x_{i,v}$. Thus, $\mathbb{E}[c(\lambda_A(v))] = 3 \ln m \cdot \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v)$. By applying the Chernoff bound with $\delta = 2$ we get that

$$\Pr[c(\lambda_A(v)) \geq 3 \cdot \mathbb{E}[c(\lambda_A(v))]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[c(\lambda_A(v))]} \leq \exp\{-\mathbb{E}[c(\lambda_A(v))]\} \leq \frac{1}{m^3}$$

where the last inequality is by using the fact that trivially, $\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v) \geq 1$ for all $v \in V$. If otherwise, then v would have no neighbors which contradicts connectivity. By applying the union bound over all n vertices we get that with probability at least $1-\frac{n}{m^3} \geq 1-\frac{1}{m}$ (assuming $m \geq 2$) for every $v \in V$, $c(\lambda_A(v)) \leq 9\ln m \cdot \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v)$. Thus, depending on the cost function, with high probability we have $\texttt{COST}_A \leq 9\ln m \cdot \sum_{v\in V}\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) = 9\ln m \cdot \texttt{OPT}$ or $\texttt{COST}_W \leq 9\ln m \cdot \max_{v\in V}\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) = 9\ln m \cdot \texttt{OPT}$. ◀

▶ **Lemma 4.** *With high probability, the above algorithm returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u)\cap\lambda_A(v)=\emptyset] &= \prod_{i\in\lambda(u)\cap\lambda(v)} \Pr[t_i > 3\ln m\cdot\min\{x_{i,u},x_{i,v}\}] \\
&= \prod_{i\in\lambda(u)\cap\lambda(v)} \{1-3\ln m\cdot\min\{x_{i,u},x_{i,v}\}\} \\
&\leq \prod_{i\in\lambda(u)\cap\lambda(v)} \exp\{-3\ln m\cdot\min\{x_{i,u},x_{i,v}\}\} \\
&= \exp\left\{-3\ln m\cdot\sum_{i\in\lambda(u)\cap\lambda(v)}\min\{x_{i,u},x_{i,v}\}\right\} \\
&\leq \exp\{-3\ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1-\frac{m}{m^3} = 1-\frac{1}{m^2}$. ◀

Combining the above lemmas gives the claim. ◀

3.3 A k-approximation algorithm for the covering problem

Algorithm 2 A relaxation-based k-approximation algorithm.

```
1 Solve the LP relaxation and obtain a fractional solution {x_{i,v}}_{i∈λ(v),v∈V}.
2 Set λ_A(v) = {i ∈ λ(v) : x_{i,v} ≥ 1/k}, for every v ∈ V.
3 return {λ_A(v)}_{v∈V}.
```

▶ **Theorem 5.** *The above algorithm is a k-approximation algorithm for both the min-sum and the min-max covering assignment problems.*

Proof. Consider an edge $uv \in E$. By averaging, there exists and interface $i \in \lambda(u)\cap\lambda(v)$ such that $\min\{x_{i,u},x_{i,v}\} \geq \frac{1}{k}$. Thus, $i \in \lambda_A(u)\cap\lambda_A(v)$ so λ_A is a covering assignment. Moreover, since $x_{i,v} \geq \frac{1}{k}$ for every $i \in \lambda_A(v)$, we have that $c(\lambda_A(v)) = \sum_{i\in\lambda_A(v)} c(i,v) \leq k\cdot\sum_{i\in\lambda_A(v)} x_{i,v}\cdot c(i,v)$. Thus, depending on the cost function, we have $\texttt{COST}_A \leq k\cdot\sum_{v\in V}\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v) = k\cdot\texttt{OPT}$ or $\texttt{COST}_W \leq k\cdot\max_{v\in V}\sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v) = k\cdot\texttt{OPT}$. ◀

3.4 ILP formulation for the connection problem

We use a similar LP to model the connection problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e we add additional flow variable y_e which is 1 if and only if the edge e is covered by the solution. Connectivity of the terminals U is enforced via cut constraints: we require that for every cut $(S, V\setminus S)$ that separates at least one pair of terminals, the

total y-weight of edges crossing the cut is at least 1. Then we can formulate the min-sum connection problem as follows:

$$\begin{aligned}
\min \quad & \sum_{v \in V} \sum_{i \in \lambda(v)} c(i,v) \cdot x_{i,v} \\
\text{s.t.} \quad & \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E \\
& y(\delta(S)) \geq 1 \quad \forall S \subset V \colon S \cap U \neq \emptyset,\ (V \setminus S) \cap U \neq \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}$$

Similarly, the min-max connection problem can be formulated as follows:

$$\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i \in \lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i \in \lambda(u) \cap \lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E \\
& y(\delta(S)) \geq 1 \quad \forall S \subset V \colon S \cap U \neq \emptyset,\ (V \setminus S) \cap U \neq \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}$$

Observe that both of the above ILPs have exponential number of constraints, but we can solve their LP relaxations in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

3.5 Randomized $O\left(\log^2 m\right)$-approximation algorithm for the connection problem

Algorithm 3 A relaxation-based $O(\log^2 m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i \in \lambda(v), v \in V}, \{y_{uv}\}_{uv \in E}$.
2. For every edge $e \in E$, let $p_e = \min\{1, 3 \ln m \cdot y_e\}$. Pick independent uniformly random variables Y_e such that $Y_e = 1$ with probability p_e and $Y_e = 0$ otherwise. Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
3. Use the $O(\log m)$-approximation algorithm for the covering problem on H to obtain a covering assignment λ_A for H.
4. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 6.** *There exists an $O\left(\log^2 m\right)$-approximation algorithm for both the min-sum and the min-max connection assignment problems which succeeds with high probability.*

Proof. We show that the algorithm for the covering problem described in the previous section can be used to obtain an $O\left(\log^2 m\right)$-approximation algorithm for the connection problem. We start with the following lemma:

▶ **Lemma 7.** *With high probability, the subgraph H connects all terminals in U.*

Proof. Fix any $S \subset V$ with $S \cap U \neq \emptyset$ and $(V \setminus S) \cap U \neq \emptyset$. Let $\ell \in \mathbb{N}$, such that $\ell \leq y(\delta(S)) < \ell + 1$. We have:

$$\begin{aligned}\Pr[|\delta_H(S)| = 0] &= \prod_{e \in \delta(S)} \Pr[Y_e = 0] = \prod_{e \in \delta(S)} (1 - p_e) \\ &\leq \prod_{e \in \delta(S)} \exp\{-p_e\} \leq \exp\left\{-\sum_{e \in \delta(S)} \min\{1, 3 \ln m \cdot y_e\}\right\} \\ &\leq \exp\{-3\ell \ln m\} = \frac{1}{m^{3\ell}}\end{aligned}$$

where the last inequality uses the LP constraint $y(\delta(S)) \geq 1$ for every U-separating cut, so $\sum_{e \in \delta(S)} y_e \geq \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [13, 12]:

▶ **Lemma 8** (Karger's lemma). *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U-separating fractional cut has a value of at least 1, so there are at most $n^{2\ell}$ U-separating cuts of value less than ℓ. By applying the union bound over all U-separating cuts we get that:

$$\begin{aligned}\Pr[\exists\, U\text{-sep. cut } S\colon |\delta_H(S)| = 0] &\leq \sum_{\ell=1}^{\infty} \sum_{\substack{S \subset V,\, S \cap U \neq \emptyset \\ (V \setminus S) \cap U \neq \emptyset,\, y(\delta(S)) < \ell+1}} \Pr[|\delta_H(S)| = 0] \\ &\leq \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{3\ell}} = \sum_{\ell=1}^{\infty} \frac{1}{m^{\ell}} = \frac{1}{m-1}\end{aligned}$$

Thus, with high probability, no U-separating cut is empty in H, i.e., H spans U. ◀

Now we want to show that by covering H we can obtain good approximation for connecting G. We have the following lemma:

▶ **Lemma 9.** *With high probability, the cost of covering H is at most $9 \ln m \cdot \mathtt{OPT}$.*

Proof. Let $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ be the fractional solution such that every variable is scaled up by a factor of $3 \ln m$. Clearly, this is a valid LP solution. Moreover, by the linearity of the LP objective function, we know that the cost of $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ is at most $3 \ln m \cdot \mathtt{OPT}$. Since for every edge $e \in E$, we have that $\Pr[Y_e = 1] = \min\{1, 3 \ln m \cdot y_e\}$, by linearity of expectations we know that the expected cost of covering H, call it $\mathtt{COST}(H)$ is at most $3 \ln m \cdot \mathtt{OPT}$. By applying the Chernoff bound with $\delta = 2$ we have that:

$$\Pr[\mathtt{COST}(H) \geq 3 \cdot \mathbb{E}[\mathtt{COST}(H)]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[\mathtt{COST}(H)]} \leq \exp\{-\mathbb{E}[\mathtt{COST}(H)]\} \leq \frac{1}{m^3}.$$

◀

Combining the above lemmas with the $O(\log m)$-approximation for the covering problem we get that with high probability, the cost of the solution is bounded by $O\big(\log^2 m\big) \cdot \mathtt{OPT}$. ◀

4 Combinatorial $O(\Delta)$-approximation algorithms

4.1 Δ-approximation for the min-sum covering problem

Algorithm 4 A simple Δ-approximation algorithm for the min-sum covering problem.

1. Initialize $\lambda_A(v) = \emptyset$ for every $v \in V$.
2. For every edge $uv \in E$, let $i_{uv} = \text{argmin}_{j \in \lambda(u) \cap \lambda(v)} \{c(j,u) + c(j,v)\}$. Set $\lambda_A(u) = \lambda_A(u) \cup \{i_{uv}\}$ and $\lambda_A(v) = \lambda_A(v) \cup \{i_{uv}\}$.
3. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 10.** *The above algorithm is a Δ-approximation algorithm for the min-sum covering problem.*

Proof. Consider any optimal solution λ_A^*. For any edge $uv \in E$, let $i_{uv}^* \in \lambda_A^*(u) \cap \lambda_A^*(v)$ be any interface that covers uv in the optimal solution. For every vertex $v \in V$ and edge $uv \in E$ incident to v, we say that v dominates uv, if $c(i_{uv}^*, v) \geq c(i_{uv}^*, u)$ (with ties broken arbitrarily). Let $D(v)$ be the set of edges dominated by v. We have:

$$\texttt{OPT} \geq \sum_{v \in V} \max_{uv \in D(v)} \{c(i_{uv}^*, u) + c(i_{uv}^*, v)\}.$$

The cost of the solution built by the procedure is upper bounded by:

$$\begin{aligned}
\sum_{uv \in E} \min_{i \in \lambda(u) \cap \lambda(v)} \{c(i,u) + c(i,v)\} &= \sum_{v \in V} \sum_{uv \in D(v)} \min_{i \in \lambda(u) \cap \lambda(v)} \{c(i,u) + c(i,v)\} \\
&\leq \sum_{v \in V} \sum_{uv \in D(v)} \max_{uv \in D(v)} \{c(i_{uv}^*, u) + c(i_{uv}^*, v)\} \\
&= \sum_{v \in V} \Delta \cdot \max_{uv \in D(v)} \{c(i_{uv}^*, u) + c(i_{uv}^*, v)\} \leq \Delta \cdot \texttt{OPT}
\end{aligned}$$

Since by construction, the solution returned by the procedure is a covering assignment, the claim follows. ◀

4.2 1.39Δ-approximation for the min-sum connection problem

Algorithm 5 A 1.39Δ-approximation algorithm for the min-sum connection problem.

1. Initialize $\lambda_A(v) = \emptyset$ for every $v \in V$.
2. For every edge $uv \in E$, let $i_{uv} = \text{argmin}_{j \in \lambda(u) \cap \lambda(v)} \{c(j,u) + c(j,v)\}$ and set $c_{uv} = c(i_{uv}, u) + c(i_{uv}, v)$.
3. Let T be a 1.39-approximate Steiner tree of G with terminal set U and edge weights c_{uv}, computed using the algorithm of [5].
4. For every edge $uv \in T$, set $\lambda_A(u) = \lambda_A(u) \cup \{i_{uv}\}$ and $\lambda_A(v) = \lambda_A(v) \cup \{i_{uv}\}$.
5. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 11.** *The above algorithm is a 1.39Δ-approximation algorithm for the min-sum connection assignment problem.*

Proof. Consider any optimal solution λ_A^* and let G^* be the subgraph of G induced by the edges covered in the optimal solution. Since λ_A^* is a connecting assignment, G^* connects all terminals in U. Let T^* be a minimum Steiner tree of G^* with terminal set U. For any edge $uv \in E(T^*)$, let $i_{uv}^* \in \lambda_A^*(u) \cap \lambda_A^*(v)$ be any interface that covers uv in T^*. For every vertex

$v \in V$ and edge $uv \in E(T^*)$ incident to v, we say that v dominates uv, if $c(i^*_{uv}, v) \geq c(i^*_{uv}, u)$ (with ties broken arbitrarily). Let $D(v)$ be the set of edges dominated by v in T^*. We have:

$$\texttt{OPT}(G) \geq \texttt{COST}_A(T^*) \geq \sum_{v \in V} \max_{uv \in D(v)} \{c(i^*_{uv}, u) + c(i^*_{uv}, v)\}.$$

Note that for every $uv \in E(T^*)$ we have $c_{uv} \leq c(i^*_{uv}, u) + c(i^*_{uv}, v)$, so $\sum_{uv \in E(T^*)} c_{uv} \leq \texttt{COST}_A(T^*) \leq \texttt{OPT}(G)$. By the 1.39-approximation guarantee of [5], the Steiner tree T satisfies $\sum_{uv \in E(T)} c_{uv} \leq 1.39 \cdot \sum_{uv \in E(T^*)} c_{uv}$. The cost of the solution is therefore at most:

$$\begin{aligned}
\sum_{uv \in E(T)} \min_{i \in \lambda(u) \cap \lambda(v)} \{c(i,u) + c(i,v)\} &\leq 1.39 \cdot \sum_{uv \in E(T^*)} \min_{i \in \lambda(u) \cap \lambda(v)} \{c(i,u) + c(i,v)\} \\
&= 1.39 \cdot \sum_{v \in V} \sum_{uv \in D(v)} \min_{i \in \lambda(u) \cap \lambda(v)} \{c(i,u) + c(i,v)\} \\
&\leq 1.39 \cdot \sum_{v \in V} \sum_{uv \in D(v)} \max_{uv \in D(v)} \{c(i^*_{uv}, u) + c(i^*_{uv}, v)\} \\
&= 1.39 \cdot \sum_{v \in V} \Delta \cdot \max_{uv \in D(v)} \{c(i^*_{uv}, u) + c(i^*_{uv}, v)\} \\
&\leq 1.39\Delta \cdot \texttt{OPT}
\end{aligned}$$

Since T is a Steiner tree spanning U and every edge of T is covered by the assignment, the solution returned by the procedure is a connecting assignment. The claim follows. ◀

4.3 2Δ-approximation for the min-max covering problem

We start with the following observation. Let M be any matching in G. Then, we have $\texttt{OPT} \geq \max_{uv \in M}\{\min_{i \in \lambda(u) \cap \lambda(v)}\{\max\{c(i,u), c(i,v)\}\}\}$. This is due to the fact, that for any edge $uv \in M$ there needs to exist a common interface $i \in \lambda_A(u) \cap \lambda_A(v)$ in the optimal solution. Since the above expression represents choosing the (localy) best interface for each edge in the matching, it is upper bounded by the cost of the optimal solution. We use the fact, that for any graph, there always exists a matching incident to every vertex of degree Δ, except at most one. We also note that if G' is a subgraph of G, then $\texttt{OPT}(G') \leq \texttt{OPT}(G)$, since any covering assignment for G is also a covering assignment for G'.

Algorithm 6 2Δ-approximation algorithm for the min-max covering problem

1. If $n = 1$ then return $\{\lambda(v) = \emptyset\}$, where v is the only vertex in G.
2. Let M be a matching in G, which is incident to the largest number of vertices of degree Δ. Let $e =$ be an arbitrary edge incident to the remaining vertex of degree Δ, if it exists.
3. For every $H \in G - (M \cup \{e\})$, call the procedure recursively and obtain a covering assignment λ_A for all vertices of H.
4. For every edge $uv \in M \cup \{e\}$, let $i_{uv} = \operatorname{argmin}_{i \in \lambda(u) \cap \lambda(v)}\{\max\{c(i,u), c(i,v)\}\}$. Set $\lambda_A(u) = \lambda_A(u) \cup \{i_{uv}\}$ and $\lambda_A(v) = \lambda_A(v) \cup \{i_{uv}\}$.
5. **return** $\{\lambda_A(v)\}_{v \in V}$.

▶ **Theorem 12.** *The above algorithm is a 2Δ-approximation algorithm for the min-max covering assignment problem.*

Proof. By construction, the solution returned by the procedure is a covering assignment. We show that its cost is at most $2\Delta \cdot \texttt{OPT}$. We claim by induction that this is the case. For

$n = 1$, i. e., $\Delta = 1$, the cost is 0 and the claim holds. Assume, that the claim holds for some value of Δ. By definition of M we get that the cost of the solution is upper bounded by:

$$\max_{uv \in M \cup \{e\}} \left\{ \min_{i \in \lambda(u) \cap \lambda(v)} \{\max\{c(i,u), c(i,v)\}\} \right\} + \max_{H \in G-(M \cup \{e\})} \mathtt{COST}_W(H)$$
$$\leq 2 \cdot \mathtt{OPT}(G) + \max_{H \in G-(M \cup \{e\})} \{2 \cdot (\Delta - 1) \cdot \mathtt{OPT}(H)\} \leq 2\Delta \cdot \mathtt{OPT}(G)$$

where the first inequality is by the fact that M and $\{e\}$ are both matchings and the induction hypothesis. ◀

5 Approximation hardness of covering and connecting multi-interface networks

The first reduction is from the Set Cover problem, which is defined as follows. We are given a universe U of n elements and a family $\mathcal{S}$ of m subsets of U. The goal is to find a subfamily $\mathcal{S}' \subseteq \mathcal{S}$ of minimum size such that $\bigcup_{S \in \mathcal{S}'} S = U$. It is well known that the Set Cover problem cannot be approximated within a factor of $o(\log n)$, unless $P = NP$ [10].

▶ **Theorem 13.** *Both the min-sum covering and connecting problems cannot be approximated within a factor of* $(1 - o(1)) \cdot \ln n$ *on stars, unless* $P = NP$.

Proof. We modify the reduction from the Set Cover problem to the min-sum covering assignment problem given by [8] which is as follows: The graph is a star with n leaves, one for each element in U. There are m interfaces, one for each set in $\mathcal{S}$. The center of the star has all m interfaces available, while the leaf corresponding to an element $u \in U$ has only those interfaces available that correspond to sets containing u. The cost of activating any interface at any leaf is 0, while the cost of activating any interface at the center is 1. It is easy to see that there is a one-to-one correspondence between the set covers of U and the covering assignments for the constructed instance. Moreover, the cost of a covering assignment is equal to the size of the corresponding set cover. Thus, if there was an $o(\log \Delta)$-approximation algorithm for the min-sum covering assignment problem on stars, then we would have an $o(\log n)$-approximation algorithm for the Set Cover problem, which contradicts the known hardness result. Since for trees the covering problem is equivalent to connecting problem the claim follows. ◀

▶ **Theorem 14.** *The min-sum covering assignment problem with unit costs cannot be approximated within a factor of* $o(\log \Delta)$ *even on bipartite graphs, unless* $P = NP$.

Proof. The reduction is similar, however instead of one center vertex, we create n copies of it, with the same set of available interfaces and we connect each leaf to all the copies of the center. The cost of activating any interface at any vertex is 1. Let v be the center representative, such that the cost $c(\lambda_A(v))$ of activating interfaces at v is smallest among all the center representatives. Firstly, we copy the assignment of v to all the center representatives we get a covering assignment, thus obtain a solution with non-increased cost. For every leaf we also leave only one interface activated, which is the one corresponding to some set covering the element represented by this leaf. We therefore have that the cost of the solution is bounded by $cn \leq \mathtt{COST} = n + n \cdot c \leq 2 \cdot n \cdot c$. Moreover, by choosing the solution to the set cover instance to be the family of sets corresponding to the interfaces activated at v we get a set cover of size at most c. Since calculating c to within a factor of 2 means calculating the size of the optimal set cover to within a factor of 2, any α-approximation algorithm for the min-sum covering assignment problem would yield an $2 \cdot \alpha$-approximation algorithm for the Set Cover problem, which implies the claim. ◀

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